# JULIO LANDA

Mechatronics Engineer
Master's Degree in Renewable Energy

## PROFESSIONAL EXPERIENCE

### NC School Cuernavaca

**Lead Physics and Mathematics Teacher** · 2024-2025

Lead teacher for physics and mathematics courses.

### IER, UNAM

**Teaching Assistant** · 2023-2025

Assisted with assignments, exams and project assessment for Introduction to Bioclimatic Design in the sixth semester of Renewable Energy Engineering, and served as teaching assistant for second-semester Linear Algebra.

**IoT Developer Engineer** · 2020 / 2024-present

Develop low-cost devices using open-source hardware and software, integrated with IoT platforms for the instrumentation project of the new Renewable Energy Engineering building at IER, UNAM.

### CNEER

**Workshop Coordinator** · 2023

Planned, organized and coordinated workshops for the 10th edition of CNEER, the National Congress of Renewable Energy Students.

### CNEER

**General Coordinator** · 2026

General Coordinator of the National Congress of Renewable Energy Students (CNEER) 2026.

### DMC Ingenieria

**Electrical Engineer** · 2022

Provided maintenance to electric motors and control panels as a contractor for Cementos Moctezuma.

### Bombas y Servicios Corbala

**Electrical Engineer** · 2021

Maintained and installed deep-well water pumps, and designed and installed control systems and electrical panels.

### Alucaps

**Maintenance Technician** · 2020

Applied maintenance plans to company machinery, updated and programmed production-line control systems using PLCs, and assembled electrical panels.

## EDUCATION

### Master's in Engineering - Energy Area

**Instituto de Energias Renovables, UNAM** · 2022-2025
Morelos, Mexico.

### Bachelor of Mechatronics Engineering

**Universidad Tecnologica de Emiliano Zapata** · 2018-2020
Emiliano Zapata, Morelos, Mexico.

### Higher University Technician in Mechatronics, Automation Area

**Universidad Tecnologica de Emiliano Zapata** · 2016-2018
Emiliano Zapata, Morelos, Mexico.

JL

## CONTACT


7771675118

landajulioc@gmail.com

JulioLanda4

Website

Priv. Cerezos No. 34, Col. Miguel Hidalgo, Jiutepec, Morelos. 62556


## LANGUAGES

- Spanish: native
- English: upper-intermediate/advanced

## SKILLS

- **Instrumentation:** instrument design and calibration.
- **Maintenance:** preventive and corrective maintenance.
- **Electrical panels:** design and assembly of electrical systems.
- **Programming:** Python, C/C++, Arduino and LabVIEW.
- **Automation:** design and development of industrial control systems.
- **Mechanical design:** CAD modeling with SolidWorks and FreeCAD.
- **Data analysis:** Python, NumPy, Pandas, SciPy, Scikit-learn, Matplotlib, Seaborn and Plotly.
- **Artificial intelligence:** machine learning, neural networks and genetic algorithms.
- **IoT:** development with ESP32/Arduino and platform integration.
- **Electronics:** circuit design and automation projects.
- **PLCs:** control system programming.

## PUBLICATIONS

### IoT smartwatch based on open technologies for the collection of thermal comfort data

**HardwareX** · 2025
DOI: https://doi.org/10.1016/j.ohx.2025.e00633

### IoT smartwatch based on open technologies for the collection of thermal comfort data

**Master's thesis** · 2025
Available at: https://altamarmx.github.io/reloj_inteligente/
Project licensed under GPL v3.0: https://github.com/JulioLanda4/IoT_Reloj_Confort/tree/v1.0.0

## CERTIFICATIONS, COURSES, HONORS & AWARDS

### Certifications

- **Certified SolidWorks Professional (CSWP)** · 2019.
- **Google IT Automation with Python** · 2022.
- **LabVIEW Associate Developer** · 2019.

### Courses

- **Macrotraining in Artificial Intelligence (MeIA)** · 2023.
- **Python for Engineering (UNAM, Coursera)** · 2023.

### Honors & Awards

- **1st place** · 2019. 3rd Mathematics Olympiad, Universidad Tecnologica de Emiliano Zapata.
- **Honorable Mention** · 2023. XXXIII CUAM-ACMor Research Congress, Physics-Mathematics category, project "Ecological Greenhouse from Plastic Waste".
- **Recognition as speaker** · 2024. Sustainable Buildings Symposium: Simulations and Digitalization for Decarbonization.
- **Honorable Mention** · 2025. XXXV CUAM-ACMor Research Congress, Physics-Mathematics category, project "Green Motion: How Do You Move?".